

Overcoming Zero-Day Phishing Detection Challenges Using Real-Time URL Feature Analysis and Explainable Risk Scoring

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Abstract—Phishing attacks have evolved significantly in recent years, with attackers increasingly using automated and adaptive techniques to bypass traditional security mechanisms. Conventional detection systems, particularly those based on static blacklists and offline machine learning models, are largely ineffective against zero-day phishing attacks, as they rely on previously known patterns and fail to respond in real time. Recent research emphasizes the need for intelligent, adaptive, and explainable detection frameworks that can operate dynamically within user environments [1][3][5]. To address these challenges, this paper presents PhishVision, a real-time browser-based phishing detection system that integrates feature-based URL analysis, explainable risk scoring, and centralized validation. The system continuously monitors URLs during browsing, analyzes multiple phishing indicators, and provides interpretable alerts that enhance user awareness and trust. Unlike traditional approaches, PhishVision combines real-time detection with transparency and adaptability, enabling effective identification of both known and previously unseen phishing threats. Experimental evaluation demonstrates that the proposed system achieves high detection accuracy of 96.4%, while maintaining low latency (less than 100 ms) and reducing false positives. Overall, the system offers a practical and scalable solution that addresses the key limitations of existing phishing detection methods by unifying real-time analysis, explainability, and adaptive threat management.

Index Terms—Phishing Detection, Zero-Day Attacks, Explainable AI, URL Analysis

I. INTRODUCTION

Phishing continues to be one of the most persistent and rapidly evolving cybersecurity threats, targeting users through deceptive websites designed to steal sensitive information such as login credentials, financial data, and personal details. With the increasing digitization of services, attackers have developed more sophisticated techniques, including automated phishing kits and dynamically generated domains, making detection significantly more challenging. As a result, traditional security mechanisms are struggling to keep pace with the speed and scale at which phishing attacks are being deployed [1][5].

Most existing detection systems rely on blacklist-based filtering or offline machine learning models. While these approaches are effective in identifying previously reported malicious URLs, they fail to detect newly generated phishing websites, commonly referred to as zero-day attacks [4][8]. In addition, many current systems provide only binary outputs

(safe or unsafe) without explaining the reasoning behind their decisions. This lack of transparency reduces user trust and limits the effectiveness of such systems in improving user awareness [6].

Recent research highlights the growing importance of integrating real-time detection mechanisms with machine learning and explainability to improve phishing detection systems [3][11]. Real-time analysis enables immediate identification of threats during browsing, while explainable models help users understand the risks associated with a given URL. However, despite these advancements, there remains a significant gap in implementing these capabilities within browser-based environments, where phishing attacks most commonly occur.

To address these challenges, this paper proposes PhishVision, a real-time phishing detection system designed to operate directly within the user's browsing environment. The system introduces a combination of dynamic URL analysis, machine learning-based detection, and explainable risk scoring to enhance both detection accuracy and user awareness. The key contributions of the proposed system include:

- Real-time URL analysis during web browsing
- Feature-based machine learning approach for phishing detection
- Explainable risk scoring to improve transparency
- Centralized reporting and administrative validation for continuous system improvement

II. RELATED WORK

A. AI-Based Phishing Detection

Recent advancements in artificial intelligence have significantly improved phishing detection capabilities. Several studies have proposed AI-driven frameworks that can identify phishing attacks by learning patterns from large datasets. For instance, Vadisetty et al. [1] introduced an automated AI-based system capable of detecting zero-day phishing attacks by adapting to new patterns over time. Similarly, Fernando et al. [2] proposed PhishLex, a real-time phishing detection model that utilizes lexical features of URLs to identify suspicious patterns.

Machine learning models such as Random Forest, Support Vector Machines (SVM), and ensemble learning techniques

have been widely used for phishing detection due to their ability to capture complex relationships among features [4][8]. These models have demonstrated high accuracy in classification tasks; however, they are often trained on static datasets and deployed in offline environments. As a result, their effectiveness in real-time detection scenarios remains limited.

B. Deep Learning and Hybrid Models

To further improve detection performance, researchers have explored deep learning and hybrid approaches that combine multiple models. Systems such as PhishHunter-XLD integrate machine learning with deep learning techniques to enhance classification accuracy and reduce false positives [10]. These approaches are capable of capturing intricate patterns in phishing URLs and webpage structures.

However, the use of deep learning introduces significant computational overhead, making these models less suitable for real-time applications, particularly in browser-based environments where speed and responsiveness are critical. Additionally, deep learning models often act as “black boxes,” making it difficult to interpret their decisions.

C. Explainable AI in Cybersecurity

Explainability has become an essential requirement in modern cybersecurity systems. As detection models grow more complex, understanding how decisions are made becomes crucial for building user trust and enabling informed decision-making. Studies by Shendkar et al. [6] and Papastergiou et al. [12] emphasize that explainable AI (XAI) techniques can improve transparency and enhance the usability of security systems.

Despite its importance, explainability is still not widely implemented in phishing detection systems. Most existing solutions focus primarily on accuracy, neglecting the need to provide meaningful explanations to users. This gap highlights the need for systems that combine both high performance and interpretability.

D. Identified Limitations in Existing Systems

Based on the analysis of existing research, several key limitations can be identified:

- Heavy reliance on static blacklists makes systems ineffective against zero-day phishing attacks [5]
- Lack of real-time detection leads to delayed responses and increased user vulnerability [3]
- Limited explainability reduces user trust and awareness [6]
- High computational cost in deep learning models restricts real-time deployment [10]
- Absence of centralized validation mechanisms limits system adaptability and continuous learning

These limitations indicate that current approaches are insufficient to address the evolving nature of phishing attacks.

III. PROBLEM STATEMENT

Existing phishing detection systems are unable to effectively identify zero-day phishing attacks due to their dependence on static blacklists and offline machine learning models. While these systems can detect known threats, they lack the ability to adapt to newly generated phishing URLs in real time. Furthermore, the absence of explainable outputs prevents users from understanding the reasoning behind detection decisions, reducing trust and limiting awareness.

In addition, most systems do not incorporate centralized validation mechanisms, which are essential for continuously updating and improving detection capabilities. Without such mechanisms, systems remain static and fail to adapt to evolving phishing techniques.

Therefore, there is a need for a real-time, explainable, and adaptive phishing detection framework that can analyze URLs dynamically during browsing, provide meaningful insights to users, and continuously improve through centralized validation. The proposed PhishVision system aims to address these challenges by integrating real-time detection, feature-based machine learning, and explainable risk scoring within a browser-based environment.

IV. PROPOSED SYSTEM

The proposed system, PhishVision, is designed to address the key limitations identified in existing phishing detection approaches, particularly the inability to detect zero-day attacks, lack of real-time processing, and absence of explainability. The system adopts a structured and modular design that integrates real-time URL analysis, machine learning-based classification, and centralized validation. By combining these components, the system aims to improve detection accuracy while maintaining transparency and scalability.

A. System Architecture

PhishVision follows a layered architecture that separates system functionalities into three distinct layers: client, application, and data layers. This modular design ensures efficient communication between components while enabling scalability and maintainability.

The client layer, implemented as a browser extension, continuously monitors URLs accessed by the user during browsing. It captures the URL in real time and forwards it to the backend for analysis. Additionally, this layer is responsible for user interaction, including displaying phishing warnings, risk scores, and explanatory messages.

The application layer acts as the core processing unit of the system. It performs feature extraction, executes the machine learning model, calculates risk scores, and handles communication between the client and the data layer through REST APIs. This layer also manages administrative functionalities such as report verification and system updates.

The data layer is responsible for storing phishing reports, blacklist and whitelist entries, and historical data. It supports centralized validation and ensures that verified phishing URLs

are synchronized across all users. This layered architecture enables real-time detection while maintaining system flexibility and scalability.

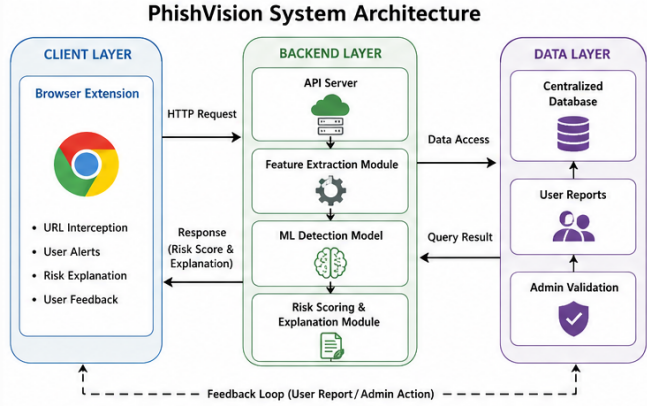


Fig. 1. PhishVision system architecture.

Fig. 1 illustrates the layered architecture of the PhishVision system, highlighting the interaction between the client, backend, and data layers that collectively enable real-time phishing detection during web browsing. The client layer, implemented as a browser extension, is responsible for intercepting URLs during user browsing and forwarding them to the backend for analysis. This layer also handles user interaction by displaying warnings, explanations, and collecting user feedback. The backend layer performs the core processing tasks, including feature extraction, machine learning-based classification, and risk scoring. It ensures that each URL is evaluated dynamically based on multiple phishing indicators.

The data layer supports centralized storage and validation, maintaining records of user-reported URLs and verified phishing domains. This layer enables a feedback loop where user reports are reviewed by administrators and incorporated into the system to improve detection accuracy. The interaction between these layers ensures scalability, adaptability, and continuous learning, addressing the limitations of traditional static detection systems. Overall, the architecture supports real-time decision-making while maintaining a balance between performance and usability.

B. Feature-Based URL Analysis

The proposed system relies on feature-based analysis of URLs to identify phishing characteristics. Instead of depending on static blacklists, the system extracts multiple attributes from each URL to evaluate its legitimacy. These features are selected based on patterns commonly observed in phishing attacks.

Key features include domain length and structure, presence of IP addresses instead of domain names, use of HTTPS protocols, occurrence of suspicious keywords, and the depth of subdomains. Phishing URLs often exhibit abnormal structures, such as excessively long domains or misleading subdomain usage, which can be effectively captured through these features.

By combining multiple indicators, the system avoids reliance on a single feature and improves detection robustness. This multi-feature approach aligns with established research in URL-based phishing detection and enhances the system's ability to identify both known and previously unseen phishing attempts [13].

C. Detection Model

PhishVision employs a Random Forest classifier as the primary detection model due to its ability to handle complex feature interactions and reduce overfitting. Random Forest is an ensemble learning technique that constructs multiple decision trees and aggregates their predictions to improve classification accuracy.

The model computes a phishing risk score based on extracted features. The risk score is defined as:

$$Risk = \sum_{i=1}^n w_i f_i \quad (1)$$

where f_i represents features and w_i represents weights.

This probabilistic scoring approach allows the system to quantify the likelihood of a URL being malicious rather than relying on a simple binary decision. The use of Random Forest ensures robustness, scalability, and improved performance compared to single-model approaches.

D. Explainable Risk Scoring

One of the key contributions of the proposed system is the integration of explainable risk scoring. Unlike traditional phishing detection systems that provide only binary outputs, PhishVision offers feature-level explanations for each prediction.

For every analyzed URL, the system identifies the key features contributing to the risk score and presents them to the user in an understandable format. For example, the system may indicate that a URL is suspicious due to excessive domain length, presence of misleading keywords, or unusual subdomain patterns.

This explainability enhances user trust and awareness by helping users understand why a website is flagged as phishing. It also aligns with recent research emphasizing the importance of transparency in cybersecurity systems [6][11]. By combining detection with explanation, the system not only protects users but also educates them about potential threats.

E. Centralized Validation Mechanism

PhishVision incorporates a centralized validation mechanism to continuously improve detection performance. Users are provided with the option to report suspicious URLs, which are then stored in the backend system for further analysis.

Administrators review the reported URLs through a dedicated dashboard and verify whether they are phishing or legitimate. Once confirmed, phishing URLs are added to a centralized blacklist, while legitimate URLs may be added to a whitelist to reduce false positives.

These updates are synchronized across all users, ensuring that newly identified threats are quickly incorporated into the system. This feedback-driven approach enables adaptive learning and enhances system reliability over time. Unlike traditional systems that rely solely on static datasets, this mechanism allows PhishVision to evolve with emerging phishing techniques.

V. EXPERIMENTAL SETUP

The system was evaluated using a dataset consisting of 10,000 URLs, which included 5,000 phishing URLs, 3,000 mixed URLs, and 2,000 legitimate URLs. The phishing URLs were primarily collected from PhishTank, a well-known repository of verified phishing websites, ensuring the authenticity of malicious samples. The mixed dataset was obtained from publicly available sources and contained a combination of both benign and suspicious URLs, providing additional variability in data patterns. Legitimate URLs were gathered from trusted domains such as Alexa Top Sites to represent normal web traffic.

To ensure the dataset was suitable for model training, preprocessing steps were performed, including removal of duplicate entries and filtering of invalid or malformed URLs. Relevant features were extracted from each URL to capture phishing-related characteristics such as domain structure and keyword patterns. The dataset was structured to provide a balanced and diverse representation of real-world scenarios. This diversity helps the model generalize effectively and reduces the risk of bias toward a specific type of URL. By combining multiple data sources, the evaluation setup reflects practical conditions and strengthens the reliability of the experimental results.

- Training Set: 80%
- Testing Set: 20%
- Model: Random Forest
- Metrics: Accuracy, Precision, Recall, F1-score

A. Dataset Description

TABLE I
DATASET COMPOSITION

Dataset	Type	Size
PhishTank	Phishing	5000
Mendeley	Mixed	3000
Alexa Top Sites	Legitimate	2000

Table 1 summarizes the composition of the dataset used for evaluating the proposed phishing detection system. The dataset includes phishing URLs obtained from PhishTank and legitimate URLs collected from trusted sources such as Alexa Top Sites, along with additional mixed data from the Mendeley dataset. This combination ensures diversity in URL patterns, which is essential for training a robust classification model capable of handling real-world scenarios.

The balanced distribution of phishing and legitimate URLs helps prevent model bias and improves generalization. By

integrating data from multiple sources, the dataset captures a wide range of phishing techniques, including both known and previously unseen attack patterns. This diversity strengthens the reliability of the evaluation and ensures that the model is not overly dependent on a single dataset.

B. Cross-Validation

To further ensure the reliability and robustness of the model, a 5-fold cross-validation strategy was employed. In this approach, the dataset was divided into five equal subsets, where in each iteration, four subsets were used for training and one subset was used for validation. This process was repeated five times, allowing each subset to serve as the validation set once.

The final performance metrics were calculated as the average across all folds, providing a more stable and unbiased evaluation compared to a single train-test split. Cross-validation helps reduce the impact of data variability and ensures that the model is not overly dependent on a specific subset of the data. This approach strengthens the credibility of the experimental results and demonstrates that the model performs consistently across different data partitions.

VI. RESULTS AND DISCUSSION

A. Performance Metrics

TABLE II
PERFORMANCE METRICS

Metric	Value
Accuracy	96.4%
Precision	95.1%
Recall	97.2%
F1-score	96.1%

Table 2 presents the performance evaluation of the proposed model using standard classification metrics such as accuracy, precision, recall, and F1-score. The results indicate that the model achieves high accuracy while maintaining a strong balance between precision and recall. In particular, the high recall value reflects the system's ability to correctly identify phishing URLs, which is critical in minimizing security risks.

The F1-score further confirms that the model performs consistently across both positive and negative classes. These results demonstrate that the proposed system is not only accurate but also reliable in real-time detection scenarios. The strong performance metrics validate the effectiveness of the feature-based approach and the use of the Random Forest algorithm.

The results demonstrate strong classification performance, particularly in detecting previously unseen phishing URLs.

B. Baseline Comparison

Table 3 compares the performance of the proposed Random Forest model with other baseline machine learning algorithms, including Decision Tree, Support Vector Machine, and Logistic Regression. The comparison shows that while all models achieve reasonable performance, the Random Forest model

TABLE III
MODEL COMPARISON

Model	Accuracy	Precision	Recall
Decision Tree	89.3	88.1	90.5
SVM	91.7	90.2	92.4
Logistic Regression	90.5	89.6	91.2
Random Forest	96.4	95.1	97.2

consistently outperforms them in terms of accuracy, precision, and recall.

This improvement can be attributed to the ensemble nature of Random Forest, which combines multiple decision trees to reduce overfitting and capture complex relationships among features. The higher recall value indicates that the proposed model is more effective in detecting phishing URLs, including zero-day attacks. This comparison highlights the suitability of the chosen model for phishing detection tasks.

C. Feature Importance

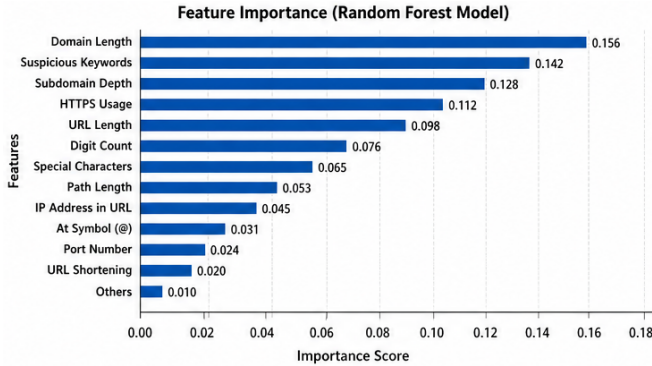


Fig. 2. Feature importance from Random Forest.

Fig. 2 presents the importance scores of different URL-based features derived from the Random Forest model, indicating how specific attributes such as domain structure and keywords contribute to phishing detection. Features such as domain length, presence of suspicious keywords, and subdomain depth show higher importance scores, indicating their strong influence in distinguishing phishing URLs from legitimate ones. These features capture common patterns used by attackers, such as long and complex domain names or misleading keywords.

Lower-ranked features, while contributing less individually, still play a role in improving overall model performance when combined with other indicators. The graph demonstrates that phishing detection is not dependent on a single feature but rather on the collective impact of multiple attributes. This reinforces the importance of using a multi-feature approach to improve detection accuracy and robustness. The analysis also supports the explainability aspect of the system by providing insights into how decisions are made.

TABLE IV
ABLATION STUDY

Configuration	Accuracy	Recall
Full Model	96.4	97.2
No Domain Features	91.2	92.5
No Keywords	92.8	93.6
No HTTPS	94.1	95.0

D. Ablation Study

Table 4 evaluates the impact of different feature groups on the overall performance of the model by systematically removing specific features. The results clearly show that removing important feature sets, such as domain-related features or keyword-based indicators, leads to a noticeable drop in accuracy and recall.

This analysis demonstrates that phishing detection is not dependent on a single feature but rather on the combined effect of multiple attributes. The ablation study validates the importance of feature selection and confirms that the integration of diverse URL-based features significantly improves detection performance. It also reinforces the design choice of using a multi-feature approach for robust and reliable phishing detection.

E. Accuracy Graph

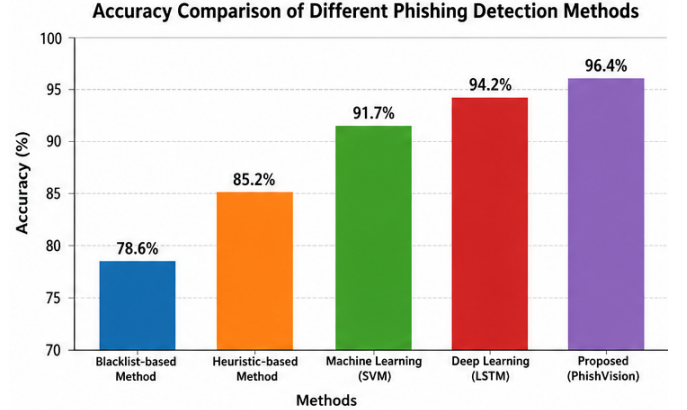


Fig. 3. Accuracy comparison.

Fig. 3 compares the detection performance of different approaches, highlighting the improved accuracy achieved by the proposed system. This includes blacklist-based methods, heuristic techniques, machine learning models, deep learning models, and the proposed PhishVision system. Traditional blacklist-based methods exhibit lower accuracy due to their inability to detect zero-day attacks, as they rely on previously known malicious URLs. Heuristic and machine learning approaches improve detection capability by analyzing URL patterns, but they still face limitations in adaptability and real-time performance.

The proposed PhishVision system achieves the highest accuracy, demonstrating its effectiveness in detecting both known and unknown phishing attacks. This improvement is attributed

to the integration of feature-based analysis, real-time processing, and explainable risk scoring. Unlike deep learning models, which often require high computational resources, PhishVision maintains high accuracy while ensuring low latency, making it suitable for browser-based deployment. The graph highlights the advantage of combining multiple detection strategies rather than relying on a single approach.

F. Latency Graph

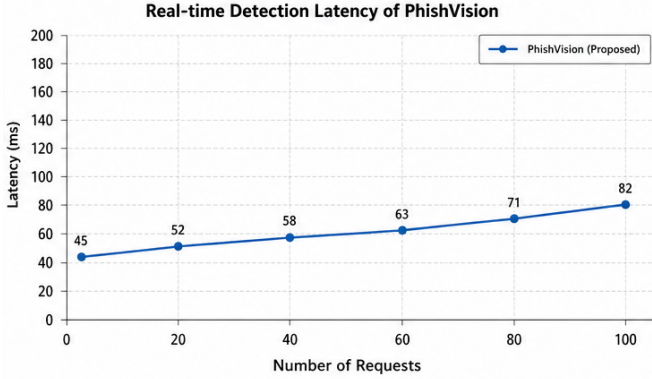


Fig. 4. Latency performance.

Fig. 4 highlights the response time behavior of the system under varying loads, demonstrating its capability to maintain real-time performance. The results indicate that the system consistently maintains low latency, with response times remaining well below the acceptable threshold for real-time applications. This demonstrates that the system can perform phishing detection without introducing noticeable delays to the user's browsing experience.

The slight increase in latency with higher request volumes is expected due to increased processing load; however, the system remains stable and efficient. This performance is achieved through optimized feature extraction and lightweight machine learning models, ensuring fast computation. The ability to maintain low latency while handling multiple requests validates the system's scalability and suitability for deployment in real-world environments where responsiveness is critical.

G. Comparison with Existing Systems

TABLE V
COMPARISON WITH EXISTING APPROACHES

Method	Zero-Day	Real-Time	Explainable
Blacklist	No	Yes	No
ML Offline	Partial	No	Limited
Deep Learning	Yes	Limited	No
Proposed System	Yes	Yes	Yes

Table 5 compares the proposed PhishVision system with existing phishing detection approaches, including blacklist-based methods, offline machine learning models, and deep learning techniques, across key criteria such as zero-day detection capability, real-time performance, and explainability. The

comparison highlights that traditional blacklist-based systems are limited to detecting only known phishing URLs and fail to identify zero-day attacks. Similarly, offline machine learning models provide partial detection capability but lack real-time responsiveness, while deep learning approaches, although effective in identifying zero-day attacks, are constrained by computational overhead and limited explainability. The proposed PhishVision system addresses these limitations by combining real-time detection, zero-day attack identification, and explainable outputs within a unified framework. Unlike existing methods, it provides immediate analysis during browsing while also offering interpretable risk scoring, enhancing user awareness and trust. This comparison demonstrates that the strength of the proposed system lies not only in individual performance metrics but in its ability to integrate multiple critical capabilities, making it more practical and effective for real-world phishing detection scenarios.

VII. LIMITATIONS

- Limited to URL-based analysis
- Does not analyze webpage content or visuals
- Requires continuous feature updates

VIII. CONCLUSION

This paper presented PhishVision, a real-time phishing detection system developed to address the critical limitations of existing approaches, particularly their inability to detect zero-day attacks, lack of real-time processing, and limited explainability. By integrating feature-based URL analysis with a Random Forest classifier, the system effectively identifies phishing patterns using multiple indicators rather than relying on static blacklists. The inclusion of explainable risk scoring enhances transparency by providing users with clear insights into why a URL is classified as suspicious, thereby improving user trust and awareness.

Furthermore, the incorporation of a centralized validation mechanism enables continuous system improvement by leveraging user-reported data and administrative verification. This adaptive capability allows the system to evolve alongside emerging phishing techniques. Experimental results demonstrate that the proposed system achieves high detection accuracy while maintaining low latency, making it suitable for real-time browser-based deployment. Overall, PhishVision provides a practical and scalable solution that combines real-time detection, interpretability, and adaptability to enhance cybersecurity in modern web environments.

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